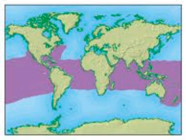
Dolichopteroides binocularis

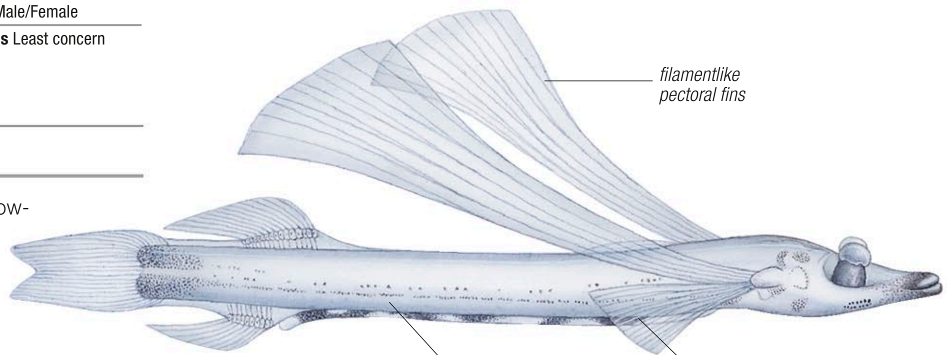
Spookfish



Length Up to 9¼ in (24 cm)
Weight Not recorded
Sex Male/Female
Status Least concern

Location Tropical and subtropical waters worldwide, W. North Atlantic

This bizarrely shaped, slow-growing fish is one of about 19 species in the spookfish family. It is distinguished from its relatives by its long, filamentlike pectoral fins, which are over half as long as the rest of its body. However, its most striking feature is the highly unusual anatomy of its eyes. Tubular in shape, they point obliquely upward, like a pair of binoculars. This would make light directed at the side of the eyes hard to detect but each eye is divided into two halves, the second half appearing as a bump on the side. These diverticular eyes give a 4-eyed appearance to the fish. Like many deep-sea fishes, the spookfish is very fragile, and prone to damage if recovered from great depths.



Plecoglossus altivelis

Ayu

Length Up to 28 in (70 cm)
Weight Not recorded
Sex Male/Female
Status Data deficient

Location W. North Pacific, E. Asia

The ayu is flattened from side to side, and has one large and one small dorsal fin and a forked tail. A vegetarian, it feeds on river algae with its modified teeth and digests food with the help of an extra-long gut. The ayu's specialized kidneys enable it to move between sea and freshwater without being affected by the changes in salinity.

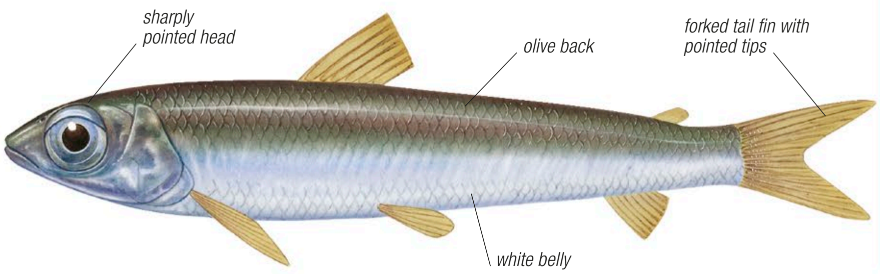
Argentina silus

Greater argentine

Length Up to 23½ in (60 cm)
Weight Over 16 oz (450 g)
Sex Male/Female
Status Not evaluated

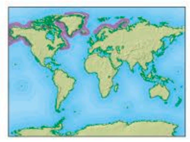
Location North Atlantic, Arctic Ocean

Argentines are marine smelts and are also known as herring smelts because they have large silvery scales with tiny spines and superficially resemble herring. The greater argentine has an olive-colored back, white belly, and silvery sides that have a golden luster. Argentines swim just above the seabed in deep water, below 492 ft (150 m), and feed on planktonic (drifting) invertebrates, such as crustaceans, squid, and comb jellies, as well as small fish. Large eyes help the fish spot and ambush prey in the dim conditions that prevail several hundred meters below the ocean surface. While little is known of its natural behavior, the greater argentine is thought to live in shoals near the seabed, which would provide some protection from the many larger fish that are likely to prey on it. The eggs laid by the female and the fry that hatch from them, float in mid-water currents between 1,313–1,640 ft (400–500 m). The greater argentine is fished for food in Europe and Russia.



Mallotus villosus

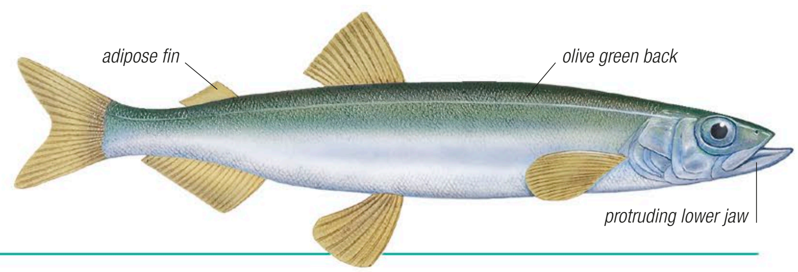
Capelin



Length Up to 8 in (20 cm)
Weight Up to 1 7/16 oz (40 g)
Sex Male/Female
Status Not evaluated

Location North Pacific, North Atlantic, Arctic Ocean

The capelin lives and breeds at sea, forming large schools that are an important source of food for seabirds, seals, dolphins, and whales; in years when capelin numbers are low, seabird rookeries may entirely fail to reproduce. Slender and with a protruding lower jaw, it is olive green on the back, merging with silver on its sides and underside, and has a large adipose fin—one of the characteristic features of salmon and their relatives. Capelin feed on small planktonic crustaceans, pelagic worms, and small fish. During the breeding season, males can be identified by the swollen base of their anal fin and by a velvety band of modified scales (villi) along the sides of their body. When courting, the male stimulates the female by caressing her with his villi, which induces her to spawn. Capelin migrate inshore to spawn on beaches, where the female lays up to 60,000 eggs at high tide. These are buried in the sand and subsequently exposed by the action of the waves. This fish is a major food source for the indigenous populations of Alaska and Canada.



Lepidogalaxias salamandroides

Dwarf pencilfish

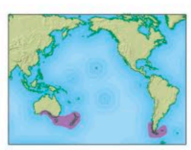
Length Up to 2¼ in (6 cm)
Weight Up to ⅞ oz (25 g)
Sex Male/Female
Status Near threatened

Location S.W. Australia

Discovered in 1961, this tiny silver-brown fish is a remarkable example of adaptation to a highly localized habitat. It is found only in the southwest corner of Western Australia, where it lives in small pools of water that lie over peaty sand. Although the water is fresh, it is black and acidic, and dries up entirely in the summer months. Dwarf pencilfishes survive during summer by burrowing up to 23½ in (60 cm) deep. To facilitate this, they have unusually flexible backbones that allow them to wriggle through the damp sand. They remain dormant for up to 5 months, breathing through their skin. Dwarf pencilfishes cannot move their eyes, because they lack external eye muscles, but they can bend their "neck" to look around. At the onset of the breeding season, males mate with females, and the eggs are fertilized inside the female's body. Each female lays up to 100 eggs, and the complete lifespan of the species is often just one year. Dwarf pencilfishes feed mainly on the larvae of aquatic insects.

Galaxias maculatus

Inanga



Length Up to 8 in (20 cm)
Weight Not recorded
Sex Male/Female
Status Least concern

Location S. Australia, New Zealand, South Pacific, S. Atlantic, S. South America

Also known as whitebait, this widespread fish of the Southern Hemisphere has a small, slender, almost cylindrical body, clear olive gray to amber above, with a blunt head, and dorsal and anal fins set back close to its tail. It has a high tolerance for sudden salinity changes, linked to its life cycle, which is divided between freshwater and the sea. The eggs are laid on estuarine plants and hatch when the next high tide occurs. The larvae spend up to 7 months at sea, before returning to mature in freshwater.